

CODING TECHNICAL ASSESSMENT

Position: Full Stack Java Developer

Level: Mid-level

Technologies: Java, HTML, CSS, JavaScript

INSTRUCTIONS:

- This is a coding-focused technical assessment
- Complete all coding exercises and projects
- Submit your code files with clear documentation
- Time limit: 2-3 hours
- Reply to this email with your completed solutions

****Full Stack Java Developer Mid-Level Coding Assessment****

****Time Limit:** 2 hours 30 minutes**

****Instructions:****

1. Please create a new folder named `full-stack-java-developer` and create separate folders for each section (coding-exercises, practical-projects, code-review-tasks).
2. Use your preferred IDE or text editor to write the code.
3. For coding exercises, please submit the code in a file with the same name as the exercise (e.g., `exercise1.java`).
4. For practical projects, please submit the entire project folder (e.g., `project1`).
5. For code review tasks, please submit the refactored code in a file with the same name as the original code (e.g., `review1.java`).
6. Submit your code as a zip file named `full-stack-java-developer.zip`.

****Section 1: CODING EXERCISES (6 questions)****

****Exercise 1: Algorithm****

* Problem Statement: Implement a binary search algorithm to find an element in a sorted array.

* Input/Output Examples:

'2 -ç WC `arr = [1, 2, 3, 4, 5, 6, 7, 8, 9]`, `target = 5`

'2 ÷WG WC `4` (index of the target element)

* Expected Time to Complete: 15 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R Æv÷ ithm

'2 Vff-6-Væ7' öb F†R Æv÷ ithm (time complexity)

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 "" Ò ³ Â "Â 2Â BÂ RÂ bÂ rÂ ,Â •Ö Â F &vWB Ò V Óâ F

'2 "" Ò ³ Â "Â 2Â BÂ RÂ bÂ rÂ ,Â •Ö Â F &vWB Ò Óâ Ó †æ÷B `ound)

Exercise 2: Data Structure

* Problem Statement: Implement a simple Stack using an array.

* Input/Output Examples:

'2 -ç WC `push(1)`, `push(2)`, `pop()`

'2 ÷WG WC `2`

* Expected Time to Complete: 20 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R 7F 0k implementation

'2 Vff-6-Væ7' öb F†R 7F 0k operations (time complexity)

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 W6,f - Â W6,f"- Â ÷ ,- Óâ &

'2 W6,f - Â W6,f"- Â W6,f2- Â ÷ ,- Óâ 6

Exercise 3: Code Implementation

* Problem Statement: Implement a simple calculator that takes in a string of mathematical operations and evaluates the result.

* Input/Output Examples:

'2 -ç WC `"2+2\*3"`

'2 ÷WG WC `8`

* Expected Time to Complete: 25 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R 6 Æ7VÆ F÷" -× ÆVÖVçF F-öà

'2 Vff-6-Væ7' öb F†R 6 Æ7VÆ F÷" ‡F-ÖR 6ö× ÆPxity)

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 #"3"£2& Óâ †

'2 # ó"Ó & Óâ F

Exercise 4: Bug Fixing

* Problem Statement: Fix the bug in the given code snippet that is supposed to reverse a string.

* Input/Output Examples:

'2 -ç WC `"hello"

'2 ÷WG WC `"olleh"

* Expected Time to Complete: 15 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R ug fix

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 &†VÆÆò& Óâ &öÆÆV,&

'2 'porld"" => `"dlrow"

Exercise 5: Code Optimization

* Problem Statement: Optimize the given code snippet that calculates the Fibonacci sequence.

* Input/Output Examples:

'2 -ç WC `n = 10`

'2 ÷WG WC `55`

* Expected Time to Complete: 20 minutes

* Evaluation Criteria:

'2 Vff-6-Væ7' öb F†R ÷ F-Ö— ed code (time complexity)

'2 6öFR V Æ—G' æB &V F &—Æ—G•

* Sample Test Cases:

'2 â Ò Óâ SV

'2 â Ò # Óâ cscV

****Exercise 6: API Development****

* Problem Statement: Implement a simple REST API using Java that exposes two endpoints: `/users` and `/users/{id}`.

* Input/Output Examples:

'2 —ç WC `GET /users`

'2 ÷WG WC `[{"id": 1, "name": "John"}, {"id": 2, "name": "Jane"}]`

* Expected Time to Complete: 30 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R ' —× ÆVÖVçF F—öà

'2 Vff—6—Væ7' öb F†R ' ‡F—ÖR 6ö× ÆPxity)

'2 6öFR V Æ—G' æB &V F &—Æ—G•

* Sample Test Cases:

'2 tUB ÷W6W'6 Óâ ·²&—B# 1, "name": "John"}, {"id": 2, "name": "Jane"}]`

'2 tUB ÷W6W'2ó Óâ ·²&—B# 1, "name": "John"}`

****Section 2: PRACTICAL PROJECTS (2 mini-projects)****

****Project 1: Todo List App****

* Problem Statement: Build a simple Todo List App using Java, HTML, CSS, and JavaScript.

* Requirements:

'2 W6W" 6 â FB æPw tasks

'2 W6W" 6 â FVÆWFR Pxisting tasks

'2 W6W" 6 â Ö k tasks as completed

* Expected Time to Complete: 45 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R —× ÆVÖVçF F—öà

'2 Vff-6-Væ7' öb F†R ‡F-ÖR 6ö× ÆPxity)

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 FB æPw task => Task is added to the list

'2 FVÆWFR Pxisting task => Task is removed from the list

'2 Ö k task as completed => Task is marked as completed

****Project 2: Weather API Client****

* Problem Statement: Build a simple Weather API Client using Java that displays the current weather for a given city.

* Requirements:

'2 W6W" 6 â -ç WB 6—G' æ ÖP

'2 F—7 Æ ys current weather for the city

'2 F—7 Æ ys error message if city not found

* Expected Time to Complete: 45 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R -× ÆVÖVçF F-öà

'2 Vff-6-Væ7' öb F†R ‡F-ÖR 6ö× ÆPxity)

'2 6öFR V Æ—G' æB &V F &-Æ—G•

* Sample Test Cases:

'2 -ç WB 6—G' æ ÖR Óâ F—7 Æ ys current weather

'2 -ç WB -àvalid city name => App displays error message

****Section 3: CODE REVIEW TASKS (2 questions)****

****Review 1: Code Refactoring****

* Problem Statement: Refactor the given code snippet to improve code quality and readability.

* Input/Output Examples:

'2 -ç WC `calculateArea(10, 20)`

'2 ÷WG WC `200`

* Expected Time to Complete: 20 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R &V`actored code

'2 6öFR V Æ—G' æB &V F &—Æ—G•

* Sample Test Cases:

'2 6 Æ7VÆ FT &V f Â# – Óâ #

'2 6 Æ7VÆ FT &V fRÂ – Óâ S

Review 2: Bug Fixing

* Problem Statement: Identify and fix the bug in the given code snippet that is supposed to calculate the sum of an array.

* Input/Output Examples:

'2 –ç WC `[1, 2, 3, 4, 5]`

'2 ÷WG WC `15`

* Expected Time to Complete: 20 minutes

* Evaluation Criteria:

'2 6÷' &V7FæW72 öb F†R ug fix

'2 6öFR V Æ—G' æB &V F &—Æ—G•

* Sample Test Cases:

'2 3 Â "Â 2Â BÂ UÖ Óâ V

'2 3 Â # Â 3 Â C Â S Ö Óâ S

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